

# Hermann Leibnitz Klaüs AGOSSOU

PhD Student | Teacher | Researcher

☎ +212 684 964 021 | ✉ hermannagossou6@gmail.com | 🔗 linkedin.com/in/agossou-hermann | 🌐 github.com/hermann-web | ✍ hermann-web.github.io/blog/blog | 📁 hermann-web.github.io/blog/projects | 📦 pypi.org/user/hermann-web | 🛂 passeport.dev

**Summary** | Math & AI PhD Student, developer and educator with hands-on experience in machine learning, software engineering, and university-level teaching. I design applied courses, mentor students, and contribute to interdisciplinary projects spanning academic inquiry and practical implementation

## Skills

- Software & System Architecture
- Full Stack Web Development
- Machine Learning & Data Science
- API Design & Development

## Tools

- Backend: Python (Django, FastAPI), Node.js
- Frontend: JavaScript/TS (React, Vue.js)
- Data Science: PyTorch, Scikit-Learn, Pandas
- DevOps: Docker, CI/CD, AWS, Kubernetes (familiar)

## Experiences

Complex Systems and Interaction Research Center (Ecole Centrale Casablanca) Sep 2024 – Present  
PhD Student in Inverse Problems for Heart Physiology Modeling (Casablanca)

- Researching inverse problems in cardiac physiology via hybrid PDE-neural network models
- Collaborating with interdisciplinary teams to advance research objectives
- Teaching and mentoring undergraduate students in data science and software engineering

Ecole Centrale Casablanca - GI Cluster - Workshop: AI Agents at the Service of Decision Jun 2025  
Lead Demo Presenter - Intelligent Decision Making using AI Agents

- Part of a 3 AI engineers team delivering a workshop to 13 participants from 5 companies including Technical Directors, R&D Directors, and technical managers.
- Personally contributed by designing and presenting an interactive Supply Chain AI Agent demo (LLMs, Streamlit).
- Developed the AI agent's core code, showcasing inventory optimisation, demand forecasting, supplier evaluation, and Return on Investment (RoI).

École Centrale Casablanca, Master's Program Mar 2025  
Course Instructor – Software Engineering Practices, March 2025 (26 hours, in-person)

Designed and delivered a practical course module on modern software engineering practices, focusing on collaborative development, code quality, and version control

- Taught principles of object-oriented programming (OOP) and software design patterns
- Guided students in implementing Model-View-Controller (MVC) architecture within data science projects
- Delivered practical workshops on Git & GitHub for version control and collaborative development
- Mentored students through a 4-day machine learning mini-project, focusing on code readability, error handling, and project organization
- Led sessions on unit testing with pytest, CI/CD principles, and dependency management in Python environments
- Evaluated students through project presentations, code quality assessment, and collaboration metrics
- Promoted best practices in code refactoring, documentation, and modular design
- Teaching Methods:
- Lectures, practical workshops, guided discussions, and group work
- Used case studies to evaluate code design and maintainability
- Delivered practical workshops on Git & GitHub for version control and collaborative development
- Focus on experiential learning and professional coding standards

Complex Systems and Interaction Research Center (Ecole Centrale Casablanca) & MELINT Jun 2023 – Jul 2024  
Research Intern with Pr. Mohammed ElRhabi: LiDAR-Based Computer Vision for Security and Surveillance (Casablanca)

- Studied LiDAR technology and its applications in security systems
- Explored and tested tools and methods for preprocessing LiDAR data, dealing with high-resolution 3D point clouds
- Fine Tuned YOLO model on lidar geometric data for detection, classification and tracking and investigated SNN
- Designed and implemented an automated scoring tool to evaluate and benchmark 17 LiDAR sensors

SIANA (JV SNCF & ONCF, High-Speed Train Maintenance) May 2022 – Sep 2022  
Data Science Research Intern with Pr. Mohammed El Rhabi: Signal Processing, TGV Maintenance (Tangier)

- Identified 4 key fault frequencies in TGV mechanics for rolling default detection
- Implemented 11 fault detection metrics in the time domain and the frequential domain (Tools: Python, Scipy.stats)
- Adapted a mutual information minimization algorithm to separate convolutive mixtures of acoustic signals (BSS)

NORMA (Legal Tech) Mar 2022 – Apr 2022  
Data Science Project Contributor: NLP Algorithms Improvement (Remote, Paris)

- Built a 6-step data retrieval process for legal documents including press releases (Tools: Transformers)
- Achieved 3x improvement in accuracy for defendant retrieval by fine-tuning BERT models (Tools: Scikit-Learn)
- Improved legal data extraction accuracy from 30% to 75%-98% for critical legal data, including monetary penalties

Safran Maroc (Aerospace, Defense)

Apr 2021 – Jul 2021

Internship: Web Development & Natural Language Processing: Knowledge Management Software (Casablanca)

- Designed, built, and deployed a full-stack knowledge management app, reducing ERP issue resolution time
- Implemented autocompletion and NLP-based search, improving search efficiency (Tools: Scikit-Learn, TF-IDF)
- Enabled solution versioning, rating, and rich-text formatting for user-submitted entries

Lycée Louis Massignon (mlfmonde/OSUI network)

Jan 2021 – Apr 2021

Math Tutor – Prépa Scientifique (MPSI), Casablanca

- Tutored MPSI students in algebra, analysis, and problem-solving; prepared and corrected exercises and mock exams
- Led twice-weekly sessions ( 64 hrs) to build core understanding and analytical skills
- Offered targeted feedback and study tips to boost performance and confidence in a competitive setting

## Education

---

University Hassan II of Casablanca & Ecole Centrale Casablanca

Casablanca, Morocco

PhD Student: Deep Hybrid Models for Cardiac Digital Twin Modelling (Physics-Informed Neural Inverse Problems)

Ecole Centrale Casablanca

Casablanca, Morocco

Engineering Degree: Digitalization & Data Science (Major), Management and Project Management (Minor) (Ranked top3)

Lycée Moulay Al Hassan

Tangier, Morocco

Preparatory Classes for Engineering Schools: Math-Physics (Ranked 2nd (1st year), 1st (2nd year))

## Certifications & Awards

---

- Advanced Machine Learning and Signal Processing - IBM (credential)
- Best Innovative Research Poster - Evaluated and awarded by Yann LeCun at TechInnov 2023 (research poster).
- Statistics for Data Science in Python - IBM (credential)
- Inferential Statistical Analysis with Python - University of Michigan (credential)
- Neural Networks and Deep Learning - deeplearning.ai (credential)

## Research / Working Papers

---

- Agossou, H., Zerhouni, K., El Kihal, C., Hamdaoui, A., Damil, N. Operator-Informed Convolutional Networks for Semi-Linear Inverse Problems: When Is Born-Inspired Filtering Enough? (Submitted, under review).  
Reproducibility archive: DOI 10.5281/zenodo.20594462
- Agossou, H., Zerhouni, K., Damil, N. Reference-State Neural Inverse Learning for Cardiac Electrophysiology Models (FitzHugh–Nagumo, Aliev–Panfilov). (Manuscript in preparation)

## Communications & Posters

---

- Should Neural Inverse Solvers Decouple Their Outputs? — Talk, JSIIA 2026, Casablanca; poster, JRC 2026, Casablanca
- Hybrid Neural Networks for Cardiac Inverse Modelling — Poster, AI for Impact 2026, SUP'RH
- On Building Efficient Neural Networks for Cardiac Digital Twins — Poster, JRECC 2025
- Neural Inverse Recovery of Cardiac Physics from Non-Invasive Measurements — Poster, JRECC 2025
- Solving Inverse Problems of Chaotic Lorenz Systems Using Neural Networks — Submitted talk, CMM 2026, Rabat
- Event-Coded Spiking Decoders for Sparse Activation-Time Inversion Under Timestamp Jitter — Submitted talk, INTIS 2026, Tangier

## Projects

---

- Agentic Quantitative Risk Analysis 🌀 Built AI agents for financial risk analysis such as a Google Finance integration, Smart Risk Detector, and Economic Evaluator, orchestrated via ARK in Kubernetes.
- Bot Detection 🌀 Built a model to classify auction bidders, reaching over 90% F1-score with linear models
- Abstract Summarization - Fine-tuned T5-small on XSum dataset for abstract summarization, achieving 1e-10 bleu
- Ozone Level Forecasting - Applied Gradient Boosting, surpassing deep learning models on meteorological data
- Summarization Software - Created an extractive summarization software using BERT/Word2Vec embeddings
- Driver Traffic Analysis 🌀 Used geospatial mapping and time series analysis to detect peak hours & traffic patterns
- Spam Detection Model - Built a machine learning model to classify emails as spam or not, achieving 95% accuracy using Random Forest and Bagging techniques.
- Event Organization - Co-organized various scientific events including the Next Generation AI Winter School (2025) 🌀, the Generative AI Winter School (2024) 🌀 and the Data Driven Decision Making Spring School (2023) 🌀
- PyPi - Developed python modules: opencf 🌀 (file conversion), flask-file-share 🌀 (file sharing web app), Common Statistics Tools 🌀, lissajou 🌀 (image and curve animations), pyv 🌀 (git branch based versioning helper), ...
- Statistical Methods Python Library (readthedocs 🌀) Implemented statistical methods, including minimum sample size, confidence interval estimation, hypothesis testing, and regression models, complete with unit test suites
- Demand Forecasting 🌀 - Built a GUI for commercial forecasting from historical data (ARIMA, LSTM, ...)

## Clubs and Associations

---

- |                                                                                      |                     |
|--------------------------------------------------------------------------------------|---------------------|
| - Centrale Coding (Programming Club) - Co-founder and mentor in machine learning     | Dec 2021 – Jun 2022 |
| - Enactus Centrale Casablanca - Community manager and Internal communication Manager | Feb 2021 – Aug 2021 |
| - Club Journal LeCentralien - Editor In Chief and Graphic Design Staff               | Jun 2020 – Jun 2021 |
| - Le Congrès - ECC (Seminar Organization) - Member of the prospection comitee        | Dec 2020 – Mar 2021 |